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GABAergic Interneuron Types in Rodent Hippocampus: Classification Schemes and Transcriptomic-Classical Mappings

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Background and Fundamental Concepts

GABAergic interneurons are inhibitory neurons that make up about 10-15% of hippocampal neurons and use GABA as their neurotransmitter to regulate excitatory activity. These cells show remarkable diversity in their shapes, functions, and molecular profiles, which has led to multiple ways to classify them over the years. (LLM Memory)

GABAergic interneurons are a diverse group of inhibitory neurons that play crucial roles in regulating hippocampal circuit function (Model-Generated). Unlike the abundant excitatory pyramidal cells, interneurons represent approximately 10-15% of all neurons in the rodent hippocampus but are essential for controlling network activity, maintaining excitatory-inhibitory balance, and generating rhythmic oscillations (Model-Generated). These cells release gamma-aminobutyric acid (GABA), the brain's primary inhibitory neurotransmitter, which hyperpolarizes target neurons and reduces their firing probability (Model-Generated). The hippocampus contains multiple subregions including the dentate gyrus, CA3, CA2, and CA1 areas, each housing distinct populations of GABAergic interneurons with specialized properties (Model-Generated). Interneuron diversity arises from differences in developmental origin, morphology, physiological properties, molecular expression profiles, and synaptic connectivity patterns (Model-Generated). This remarkable heterogeneity has made interneuron classification a central challenge in neuroscience, with researchers developing various schemes based on anatomical features, electrophysiological characteristics, molecular markers, and more recently, transcriptomic profiles (Model-Generated). Understanding interneuron types is critical because different subtypes perform distinct computational functions, from providing perisomatic inhibition that controls spike timing to dendritic inhibition that regulates synaptic integration (Model-Generated).

Classification Schemes and Methodological Approaches

Researchers use multiple approaches to classify interneurons including morphology, molecular markers, electrophysiology, and connectivity patterns, with estimates ranging from 16 to over 28 different types in the hippocampus. Recent advances in single-cell RNA sequencing have added transcriptomic classification as a powerful new method that researchers are now working to map onto traditional classification schemes. (9 sources)

The classification of GABAergic interneurons has evolved through multiple methodological approaches, each revealing different aspects of interneuron diversity. Traditional morphological classification schemes have

identified 16 or more interneuron types in the hippocampus based on anatomical features (Contreras et al., 2019), with some estimates reaching at least 26 different types (Maynard et al., 2020) or even 28 previously described types in the CA1 region alone (Hewitt et al., 2021). The field has historically used a three-part approach combining morphological, biochemical, and physiological properties to characterize interneuron subtypes (Hewitt et al., 2021) (DeFelipe et al., 2013).

Molecular classification schemes rely on the expression of specific markers, with calcium-binding proteins and neuropeptides serving as key distinguishing features (Huang et al., 2014). Parvalbumin (PV) and cholecystokinin (CCK) represent the two major populations of GABAergic neurons, with their relative distributions varying across brain regions (Contreras et al., 2019) (Whissell et al., 2015). Within the PV-expressing population alone, researchers have identified multiple subtypes including basket cells, axo-axonic cells, and bistratified cells, which can be further subdivided based on physiological properties into fast-spiking and non-fast-spiking variants (Ekins et al., 2020) (Dannenberg et al., 2017).

The advent of single-cell RNA sequencing has revolutionized interneuron classification by providing transcriptomic profiles that reveal enormous genomic diversity (Ekins et al., 2020) (Lim et al., 2018). Current research efforts focus on correlating these transcriptomic datasets with previously identified interneuron subtypes based on morphological and physiological criteria (Ekins et al., 2020). This integration of multiple classification approaches - morphology, connectivity, firing patterns, and gene expression - provides a more comprehensive framework for understanding interneuron diversity (Contreras et al., 2019) (Dannenberg et al., 2017).

Evidence

(Contreras et al., 2019)

Molecular Specialization of GABAergic Synapses on the Soma and Axon in Cortical and Hippocampal Circuit Function and Dysfunction

"Interneurons can be classified by their morphology, connectivity, firing pattern, and gene expression pattern. Based on morphology, 16 or more types of interneuron have been distinguished in the hippocampus (Cajal, 1893;Lorente and De Nó, 1934;(Parra et al., 1998) while transcriptomic cell typing has identified 23 interneuron types in the cortex (Tasic et al., 2016)"

"the majority of interneurons in these regions express either the neuropeptide cholecystokinin or the calcium binding protein parvalbumin (CCK; PV; (Whissell et al., 2015)."

(Maynard et al., 2020)

TrkB Signaling Influences Gene Expression in Cortistatin-Expressing Interneurons

"At least 26 different types of GABAergic interneurons have been identified in the hippocampus (Somogyi et al., 2004), and perhaps more in the cerebral cortex (Myers et al., 2007)(Habib et al., 2017). Differences in firing properties, connectivity patterns, and molecular expression profiles are hypothesized to contribute to nonoverlapping functions of the respective classes."

(Hewitt et al., 2021)

High and low expression of the hyperpolarization activated current (I h) in mouse CA1 stratum oriens interneurons

"The elusive classification of the vast number of GABAergic interneuron subtypes has captivated neuroscientists for decades"

"Confident classification of the subtype of GABAergic interneurons is necessary in order to understand the particular role an interneuron plays in the local circuitry. Dissection of the functional roles of different inhibitory neurons has been historically difficult. Many studies utilize a three-part approach that entail describing the morphological, biochemical, and physiological properties (Ascoli et al., 2008)(DeFelipe et al., 2013). Within the hippocampus, the CA1 region alone contains at least 28 previously described types of inhibitory interneurons located throughout all layers: stratum oriens, stratum pyramidale, stratum radiatum, and stratum lacunosummoleculare (Klausberger & Somogyi, 2008)."

(DeFelipe et al., 2013)

New insights into the classification and nomenclature of cortical GABAergic interneurons

"A systematic classification and accepted nomenclature of neuron types is much needed but is currently lacking. This article describes a possible taxonomical solution for classifying GABAergic interneurons of the cerebral cortex based on a novel, web-

based interactive system that allows experts to classify neurons with pre-determined criteria. Using Bayesian analysis and clustering algorithms on the resulting data, we investigated the suitability of several anatomical terms and neuron names for cortical GABAergic interneurons. Moreover, we show that supervised classification models could automatically categorize interneurons in agreement with experts' assignments. These results demonstrate a practical and objective approach to the naming, characterization and classification of neurons based on community consensus."

(Huang et al., 2014)

Cell type-specific expression of Eps8 in the mouse hippocampus

"With respect to the calcium-binding proteins and peptides that characterize different subpopulations of GABAergic interneurons [13](Gulyás et al., 1992)(Freund et al., 1998), we found substantial co-expression of Eps8 and calbindin, but not parvalbumin or calretinin. We also found that Eps8 is present in a subset of cholecystokinin-expressing interneurons. Taken these findings into consideration, our results suggest that Eps8expressing cells in hippocampal area CA1 predominantly belong to the calbindin-and cholecystokinin-expressing subpopulations of GABAergic interneurons. In addition to using molecular expression profiles, GABAergic interneurons can be distinguished on the basis of their morphology, electrophysiological features, and innervations of distinct subcellular domains of pyramidal cells (DeFelipe, 1997)(Somogyi et al., 1998). So far, more than 20 different types of interneurons have been recognized in the hippocampus and neocortex (Bartolini et al., 2013). It has been reported previously that CA1 pyramidal cells are supported by at least 16 distinct types of GABAergic interneurons, each having highly stereotypic laminar arrangements and functions (Somogyi et al., 2005)(Klausberger et al., 2008)."

(Whissell et al., 2015)

Comparative density of CCK- and PV-GABA cells within the cortex and hippocampus

"Cholecystokinin (CCK)- and parvalbumin (PV)-expressing neurons constitute the two major populations of perisomatic GABAergic neurons in the cortex and the hippocampus. As CCK- and PV-GABA neurons differ in an array of morphological, biochemical and electrophysiological features, it has been proposed that they form distinct inhibitory ensembles which differentially contribute to network oscillations and behavior. However, the relationship and balance between CCK- and PV-GABA neurons in the inhibitory networks of the brain is currently unclear as the distribution of these cells has never been compared on a large scale. Here, we systemically investigated the distribution of CCK- and PV-GABA cells across a wide number of discrete forebrain regions using an intersectional genetic approach. Our analysis revealed several novel trends in the distribution of these cells. While PV-GABA cells were more abundant overall, CCK-GABA cells outnumbered PV-GABA cells in several subregions of the hippocampus, medial prefrontal cortex and ventrolateral temporal cortex. Interestingly, CCK-GABA cells were relatively more abundant in secondary/ association areas of the cortex (V2, S2, M2, and AudD/AudV) than they were in corresponding primary areas (V1, S1, M1, and Aud1). The reverse trend was observed for PV-GABA cells. Our findings suggest that the balance between CCK- and PV-GABA cells in a given cortical region is related to the type of processing that area performs; inhibitory networks in the secondary cortex tend to favor the inclusion of CCK-GABA cells more than networks in the primary cortex. The intersectional genetic labeling approach employed in the current study expands upon the ability to study molecularly defined subsets of GABAergic neurons. This technique can be applied to the investigation of neuropathologies which involve disruptions to the GABAergic system, including schizophrenia, stress, maternal immune activation and autism."

(Ekins et al., 2020)

Emergence of non-canonical parvalbumin-containing interneurons in hippocampus of a murine model of type I lissencephaly

"Inhibitory interneurons are classified based on a combination of their morphological, biochemical, intrinsic electrical, and connectivity properties (Lim et al., 2018). Advances in single-cell RNA sequencing have revealed enormous diversity in interneuron genomics, and current efforts attempt to correlate transcriptomic data sets with previously identified interneuron subtypes (Tasic et al., 2018)Muñoz-Manchado et al., 2018;(Gouwens et al., 2019)Que and Lukacsovich, 2020). In CA1 hippocampus alone, inhibitory synaptic transmission is mediated by at least 15 different subtypes of GABAergic inhibitory interneurons . Three canonical interneuron subtypes express the calcium-binding protein parvalbumin (PV): basket-cells, axo-axonic cells, and bistratified cells"

"WT PV+INTs consist of two physiological subtypes (80% fast-spiking (FS), 20% non-fast-spiking (NFS)) and four morphological subtypes (basket, axo-axonic, bistratified, radiatum-targeting)."

(Dannenberg et al., 2017)

Modulation of Hippocampal Circuits by Muscarinic and Nicotinic Receptors

"Interneurons display an immense functional, morphological and genetic diversity, as described in previous reviews of the classification of interneuron subtypes (McBain et al., 2001)(Somogyi et al., 2005)(Rudy et al., 2011)(Kepecs et al., 2014)"

"Distinct expression profiles of neuropeptides that relate to points of synaptic contact, morphological structure and intrinsic excitability (Freund et al., 1998)(Kepecs et al., 2014) distinguish interneurons"

"Interneurons expressing the calcium binding protein parvalbumin (PV) make up approximately 40% of all GABAergic interneurons. However, this is a heterogeneous group of functionally distinct interneuron subtypes. For example, in the hippocampus alone, there are at least three functionally and morphologically distinct populations of PV + expressing interneurons, namely basket, axo-axonic

and bistratified cells. Fast spiking interneurons in the hippocampus and neocortex are often PV + positive and target the soma of pyramidal cells."

(Lim et al., 2018)

Development and Functional Diversification of Cortical Interneurons.

"In the cerebral cortex, GABAergic interneurons have evolved as a highly heterogeneous collection of cell types that are characterized by their unique spatial and temporal capabilities to influence neuronal circuits. Current estimates suggest that up to 50 different types of GABAergic neurons may populate the cerebral cortex, all derived from progenitor cells in the subpallium, the ventral aspect of the embryonic telencephalon. In this review, we provide an overview of the mechanisms underlying the generation of the distinct types of interneurons and their integration in cortical circuits. Interneuron diversity seems to emerge through the implementation of cell-intrinsic genetic programs in progenitor cells, which unfold over a protracted period of time until interneurons acquire mature characteristics. The developmental trajectory of interneurons is also modulated by activity-dependent, non-cell-autonomous mechanisms that influence their ability to integrate in nascent circuits and sculpt their final distribution in the adult cerebral cortex."

Classical Functional and Morphological Interneuron Types

The hippocampus contains 21 or more distinct types of GABAergic interneurons that can be classified based on their morphology, firing patterns, molecular markers, and target domains. Major types include basket cells (both fast-spiking parvalbumin-positive and regular-spiking CCK-positive), axo-axonic cells, bistratified cells, oriens-lacunosum-moleculare cells, and neurogliaform/ivy cells. (21 sources)

Based on morphological, physiological, and molecular characteristics, researchers have identified 21 different subtypes of GABAergic interneurons in the mammalian hippocampus (Molgaard et al., 2014) (Fasano et al., 2017), with some studies identifying at least 13 distinct morpho-physiological types in the dentate gyrus alone (Degro et al., 2022). The major classical interneuron types include:

Basket Cells: Two functionally distinct types target the perisomatic region of pyramidal cells. Fast-spiking parvalbumin-positive (PV+) basket cells provide rapid, high-frequency inhibition and can fire bursts up to 150 Hz during ripples (Rivera et al., 2014) (Cserep et al., 2017) (Klausberger et al., 2005). Regular-spiking cholecystokinin-positive (CCK+) basket cells fire at lower frequencies (2-10 Hz on average) and express cannabinoid CB1 receptors (Fasano et al., 2017) (Cserep et al., 2017) (Klausberger et al., 2005) (Katona et al., 1999). These two basket cell populations account for approximately 54.5% (PV+) and 45% (CB1R+/CCK+) of all GABAergic boutons on pyramidal cell somata (Cserep et al., 2017).

Axo-axonic Cells: These PV+ interneurons specifically target the axon initial segment of pyramidal cells, controlling spike initiation (Rivera et al., 2014) (Chamberland et al., 2012).

Bistratified Cells: Regular-spiking interneurons that provide dendritic inhibition and contribute to feedforward inhibition of CA1 pyramidal cells (Ferrante et al., 2015) (Bocchio et al., 2024).

Oriens-Lacunosum-Moleculare (O-LM) Cells: Dendrite-targeting interneurons that regulate dendritic spikes and synaptic plasticity by innervating the distal dendritic domains of pyramidal neurons (Weng et al., 2010) (Kraushaar et al., 2000). Two main types of somatostatin-containing cells in CA1 are O-LM cells and hippocampo-septal (HS) cells (Takacs et al., 2024).

Neurogliaform and Ivy Cells: These constitute one of the largest hippocampal interneuron populations, expressing neuronal nitric oxide synthase (nNOS) and providing slow, diffuse inhibition (Tricoire et al., 2010) (Bocchio et al., 2024). Despite apparent morphological differences, ivy cells and nNOS+ neurogliaform cells share overlapping developmental, electrophysiological, and neurochemical properties (Tricoire et al., 2010).

Interneuron-Selective Interneurons: Specialized cells that selectively target other interneurons rather than pyramidal cells. These include calretinin (CR)+ or vasoactive intestinal polypeptide (VIP)+ cells that can be subdivided into three subtypes with distinct anatomical and neurochemical features (Tyan et al., 2014) (Bocchio et al., 2024).

Region-Specific Types: The dentate gyrus contains additional specialized types, while CA1 includes specific

subtypes like outer molecular layer (OML) interneurons that project to the subiculum and can fire at high frequencies up to 130 Hz (Ceranik et al., 1997). Functional classification schemes have also identified interneuron types based on metabotropic glutamate receptor responses, distinguishing four major subtypes with different current responses and molecular markers (Hooft et al., 2000).

The complexity of interneuron classification is reflected in studies showing that morphological and physiological properties can be combined to identify five distinct classes, with some cell types like molecular layer cells being distinguished primarily by electrophysiological characteristics (Hosp et al., 2013) (Mott et al., 1997). At the molecular level, these classical types are defined by expression of calcium-binding proteins (parvalbumin, calretinin, calbindin), neuropeptides (somatostatin, NPY, CCK), and enzymes like neuronal nitric oxide synthase (Perrenoud et al., 2022) (Tepper et al., 2010).

Evidence

(Molgaard et al., 2014)

Immunofluorescent visualization of mouse interneuron subtypes

"In the mammalian hippocampus, these are divided into 21 different subtypes of GABAergic interneurons based on their expression of different markers, morphology and their electrophysiological properties."

(Fasano et al., 2017)

Regulation of the Hippocampal Network by VGLUT3-Positive CCK- GABAergic Basket Cells

"different classes of interneurons interact with specific cellular domains of pyramidal cells to modulate cell firing, plasticity and network activity. These interneurons form a complex group composed of up to 21 types of cells displaying a variety of molecular expression profiles and distinct morphological and physiological features (Somogyi et al., 2003)(Klausberger et al., 2008). Basket cells are interneurons that form a dense plexus of inhibitory terminals on the soma of pyramidal cells. Two types of basket cells have been identified: the classic parvalbumin-positive (PV) fast-firing interneurons and the regular-firing cholecystokininexpressing (CCK) interneurons."

(Degro et al., 2022)

Interneuron diversity in the rat dentate gyrus: An unbiased in vitro classification

"Information processing in cortical circuits, including the hippocampus, relies on the dynamic control of neuronal activity by GABAergic interneurons (INs). INs form a heterogenous population with defined types displaying distinct morphological, molecular, and physiological characteristics. In the major input region of the hippocampus, the dentate gyrus (DG), a number of IN types have been described which provide synaptic inhibition to distinct compartments of excitatory principal cells (PrCs) and other INs"

"This analysis reveals that DG INs divide into at least 13 distinct morpho-physiological types which reflect the complexity of the local IN network and serve as a basis for further network analyses."

(Rivera et al., 2014)

Localization of the cannabinoid CB1 receptor and the 2-AG synthesizing (DAGLα) and degrading (MAGL, FAAH) enzymes in cells expressing the Ca2+-binding proteins calbindin, calretinin, and parvalbumin in the adult rat hippocampus

"Parvalbumin + cells were specifically localized in the granular and polymorphic cell layers of the dentate gyrus and the strata oriens and pyramidale in CA1/3 fields of the rat hippocampus (Kosaka et al., 1987). They have been considered a subpopulation of GABAergic interneurons, including basket and axo-axonic cell types, which innervate the somata and proximal axons of pyramidal cells, respectively (Soriano et al., 1990)"

"Most CB + 1 terminals surrounding the somata and proximal dendrites of pyramidal neurons were cholecystokinin + (CCK) GABAergic interneurons (basket cells) and, to a lower extent, calbindin D-28k + GABAergic interneurons (Katona et al., 1999) (Marsicano et al., 1999)(Tsou et al., 1999). However, parvalbumin + GABAergic interneuron terminals localized in pyramidal cell layers were negative for CB 1 (Katona et al., 1999)(Marsicano et al., 1999)"

"most fiber terminals containing CB 1 /CCK probably innervate both calbindin + pyramidal neurons (probably including calbindin + / FAAH + pyramidal neurons in CA1) and calretinin + and parvalbumin + interneurons localized in the principal layers of the hippocampus (Gulyás et al., 1991)(Marsicano et al., 1999). Moreover, parvalbumin + fiber terminals, which represent another GABAergic interneuron type, could specifically innervate FAAH + /calbindin − pyramidal neurons in CA3."

(Cserep et al., 2017)

Mitochondrial Ultrastructure Is Coupled to Synaptic Performance at Axonal Release Sites

"In the hippocampal CA1 area, the perisomatic GABAergic inputs of principal cells can be divided into two, morphologically and functionally well-characterized populations: the CB1R□ regular-spiking interneurons show low level of synaptic activity, whereas the PV□ fast-spiking interneurons are capable of performing exceptionally high synaptic activity. In fact, these GABAergic populations are the two well-known archetypes of the slow/modulatory, and the highly energized/fast "clockwork" type interneurons, respectively (Klausberger et al., 2005)Gulyás et al., 2006;(Klausberger et al., 2005). The average in vivo firing frequency of the CB1R□ interneurons in the hippocampus is 2-10 Hz, whereas the PV□ interneurons can fire bursts up to 150 Hz during ripples (Klausberger et al., 2005) (Lapray et al., 2012)"

"54.5% of all GABAergic boutons on pyramidal cell somata were PV□ ,45% CB1R□ ,and only 0.5% were double-negative (n □ 200 boutons in two mice)"

(Klausberger et al., 2005)

Complementary Roles of Cholecystokinin- and Parvalbumin-Expressing GABAergic Neurons in Hippocampal Network Oscillations

"In the hippocampal CA1 area, a relatively homogenous population of pyramidal cells is accompanied by a diversity of GABAergic interneurons. Previously, we found that parvalbumin-expressing basket, axo-axonic, bistratified, and oriens-lacunosum moleculare cells, innervating different domains of pyramidal cells, have distinct firing patterns during network oscillations in vivo. A second family of interneurons, expressing cholecystokinin but not parvalbumin, is known to target the same domains of pyramidal cells as do the parvalbumin cells. To test the temporal activity of these independent and parallel GABAergic inputs, we recorded the precise spike timing of identified cholecystokinin interneurons during hippocampal network oscillations in anesthetized rats and determined their molecular expression profiles and synaptic targets. The cells were cannabinoid receptor type 1 immunopositive. Contrary to the stereotyped firing of parvalbumin interneurons, cholecystokinin-expressing basket and dendrite-innervating cells discharge, on average, with 1.7 ± 2.0 Hz during high-frequency ripple oscillations in an episode-dependent manner. During theta oscillations, cholecystokinin-expressing interneurons fire with 8.8 ± 3.3 Hz at a characteristic time on the ascending phase of theta waves (155 ± 81°), when place cells start firing in freely moving animals. The firing patterns of some interneurons recorded in drug-free behaving rats were similar to cholecystokinin cells in anesthetized animals. Our results demonstrate that cholecystokinin- and parvalbumin-expressing interneurons make different contributions to network oscillations and play distinct roles in different brain states. We suggest that the specific spike timing of cholecystokinin interneurons and their sensitivity to endocannabinoids might contribute to differentiate subgroups of pyramidal cells forming neuronal assemblies, whereas parvalbumin interneurons contribute to synchronizing the entire network."

(Katona et al., 1999)

Presynaptically Located CB1 Cannabinoid Receptors Regulate GABA Release from Axon Terminals of Specific Hippocampal Interneurons

"To understand the functional significance and mechanisms of action in the CNS of endogenous and exogenous cannabinoids, it is crucial to identify the neural elements that serve as the structural substrate of these actions. We used a recently developed antibody against the CB1 cannabinoid receptor to study this question in hippocampal networks. Interneurons with features typical of basket cells showed a selective, intense staining for CB1 in all hippocampal subfields and layers. Most of them (85.6%) contained cholecystokinin (CCK), which corresponded to 96.9% of all CCK-positive interneurons, whereas only 4.6% of the parvalbumin (PV)-containing basket cells expressed CB1. Accordingly, electron microscopy revealed that CB1-immunoreactive axon terminals of CCK-containing basket cells surrounded the somata and proximal dendrites of pyramidal neurons, whereas PV-positive basket cell terminals in similar locations were negative for CB1. The synthetic cannabinoid agonist WIN 55,212-2 (0.01–3 μm) reduced dose-dependently the electrical field stimulation-induced [3H]GABA release from superfused hippocampal slices, with an EC50 value of 0.041 μm. Inhibition of GABA release by WIN 55,212-2 was not mediated by inhibition of glutamatergic transmission because the WIN 55,212-2 effect was not reduced by the glutamate blockers AP5 and CNQX. In contrast, the CB1 cannabinoid receptor antagonist SR 141716A (1 μm) prevented this effect, whereas by itself it did not change the outflow of [3H]GABA. These results suggest that cannabinoid-mediated modulation of hippocampal interneuron networks operate largely via presynaptic receptors on CCK-immunoreactive basket cell terminals. Reduction of GABA release from these terminals is the likely mechanism by which both endogenous and exogenous CB1 ligands interfere with hippocampal network oscillations and associated cognitive functions."

(Chamberland et al., 2012)

Inhibitory control of hippocampal inhibitory neurons

"The population of GABAergic interneurons is extremely heterogeneous and comprises, in many brain regions, cells with divergent morphological and physiological properties, distinct molecular expression profiles, and highly specialized functions"

"Different types of hippocampal inhibitory interneurons control spike initiation [e.g., axo-axonic and basket cells (BCs)] and synaptic integration (e.g., bistratified and oriens–lacunosum moleculare interneurons) within pyramidal neurons"

(Ferrante et al., 2015)

Distinct and synergistic feedforward inhibition of pyramidal cells by basket and bistratified interneurons

"FFI in CA1PCs is mediated by two physiologically and morphologically distinct GABAergic interneurons: fast-spiking, perisomatic-

targeting basket cells and regular-spiking, dendritic-targeting bistratified cells."

(Bocchio et al., 2024)

Functional networks of inhibitory neurons orchestrate synchrony in the hippocampus

"This mouse line allows for a large sampling from diverse interneuron subtypes in the CA1 pyramidal layer, including the most representative ones (Bezaire et al., 2013), the PV-expressing basket and bistratified cells (Klausberger et al., 2008), the NOS-expressing ivy cells (14381729) and two types of interneuron-selective interneurons (ISI 1 and 3) that express calretinin (Klausberger et al., 2008)(Topolnik et al., 2022)"

(Weng et al., 2010)

Cell Type-Specific Expression of Acid-Sensing Ion Channels in Hippocampal Interneurons

"At least two functionally distinct classes of GABAergic interneurons are known to exist in the hippocampus (McBain et al., 2001)(Jonas et al., 2004)(Freund et al., 2007). Soma-targeting inhibitory interneurons, such as fast-spiking basket cells (BCs), control the spike initiation of principal neurons via axonal innervations onto perisomatic areas of principal neurons (Cobb et al., 1995)(Miles et al., 1996)(Kraushaar et al., 2000), whereas dendrite-targeting inhibitory interneurons, such as oriens lacunosum-moleculare (O-LM) cells, regulate dendritic Na ⁺ or Ca 2⁺ spikes and synaptic plasticity by innervating dendritic domains of principal neurons (Miles et al., 1996). These two distinct GABAergic interneuron subtypes differ not only in their intrinsic properties, such as neurochemical contents, Ca 2⁺ bufferingproteins/homeostasis, ion channels, and transmitter receptors (Koh et al., 1995)Freund and Buzsa ´ki, 1996;(Martina et al., 1998)(Lien et al., 2002)(Pouille et al., 2004)(Aponte et al., 2008)(Liao et al., 2009), but also in their synaptic and network functions (Klausberger et al., 2003)(Pouille et al., 2004)Klausberger and Somogyi, 2008)."

(Kraushaar et al., 2000)

Efficacy and Stability of Quantal GABA Release at a Hippocampal Interneuron–Principal Neuron Synapse

"We have examined factors that determine the strength and dynamics of GABAergic synapses between interneurons [dentate gyrus basket cells (BCs)] and principal neurons [dentate gyrus granule cells (GCs)] using paired recordings in rat hippocampal slices at 34°C. Unitary IPSCs recorded from BC–GC pairs in high intracellular Cl− concentration showed a fast rise and a biexponential decay, with mean time constants of 2 and 9 msec. The mean quantal conductance change, determined directly at reduced extracellular Ca2+/Mg2+concentration ratios, was 1.7 nS. Quantal release at the BC–GC synapse occurred with short delay and was highly synchronized. Analysis of IPSC peak amplitudes and numbers of failures by multiple probability compound binomial analysis indicated that synaptic transmission at the BC–GC synapse involves three to seven release sites, each of which releases transmitter with high probability (≥ 0.5in 2 mmCa2+/1 mm Mg2+). Unitary BC–GC IPSCs showed paired-pulse depression (PPD); maximal depression, measured for 10 msec intervals, was 37%, and recovery from depression occurred with a time constant of 2 sec. Paired-pulse depression was mainly presynaptic in origin but appeared to be independent of previous release. Synaptic transmission at the BC–GC synapse showed frequency-dependent depression, with half-maximal decrease at 5 Hz after a series of 1000 presynaptic action potentials. The relative stability of transmission at the BC–GC synapse is consistent with a model in which an activity-dependent gating mechanism reduces release probability and thereby prevents depletion of the releasable pool of synaptic vesicles. Thus several mechanisms converge on the generation of powerful and sustained transmission at interneuron–principal neuron synapses in hippocampal circuits."

(Takacs et al., 2024)

Synaptic and dendritic architecture of different types of hippocampal somatostatin interneurons

"GABAergic inhibitory neurons fundamentally shape the activity and plasticity of cortical circuits. A major subset of these neurons contains somatostatin (SOM); these cells play crucial roles in neuroplasticity, learning and memory in many brain areas including the hippocampus, and are implicated in several neuropsychiatric diseases and neurodegenerative disorders. Two main types of SOM-containing cells in area CA1 of the hippocampus are oriens-lacunosum-moleculare (OLM) cells and hippocampo-septal (HS) cells."

(Tricoire et al., 2010)

Common Origins of Hippocampal Ivy and Nitric Oxide Synthase Expressing Neurogliaform Cells

"GABAergic interneurons critically regulate cortical computation through exquisite spatiotemporal control over excitatory networks. Precision of this inhibitory control requires a remarkable diversity within interneuron populations that is largely specified during embryogenesis. Although interneurons expressing the neuronal isoform of nitric oxide synthase (nNOS) constitute the largest hippocampal interneuron cohort their origin and specification remain unknown. Thus, as neurogliaform cells (NGC) and Ivy cells (IvC) represent the main nNOS+ interneurons, we investigated their developmental origins"

"GABAergic neurons expressing nNOS are one of the largest interneuron populations in the hippocampus (Jinno et al., 2002)(B et al., 2005)(52852755)"

"IvCs and nNOS ⁺ NGCs have completely overlapping developmental, electrophysiological, morphological, and neurochemical properties, suggesting that these cells constitute a single unique interneuron subtype despite differences in laminar organization. Surprisingly, classically defined NGCs can be subdivided into two groups based on nNOS expression and lineage with nNOS ⁺ NGCs

being derived from the MGE and nNOS □ NGCs arising from CGE progenitors"

"IvCs and NGCs have been argued to represent distinct interneuron subtypes (52852755), despite the rich similarity between these cell types, including coexpression of NPY with nNOS, dense axonal arbors, and slow pyramidal cell inhibition. We found that both of these interneurons exhibit a LS phenotype and fail to express other classical interneuron markers such as PV, SOM, or CR."

(Tyan et al., 2014)

Dendritic Inhibition Provided by Interneuron-Specific Cells Controls the Firing Rate and Timing of the Hippocampal Feedback Inhibitory Circuitry

"The so-called interneuron-specific (IS) cells were identified based on direct ultrastructural evidence that some calretinin (CR)-expressing or vasoactive intestinal polypeptide (VIP)-expressing GABAergic cells in the CA1 area of the hippocampus contact interneurons selectively. IS cells were further subdivided into three subtypes with distinct anatomical and neurochemical features."

(Ceranik et al., 1997)

A Novel Type of GABAergic Interneuron Connecting the Input and the Output Regions of the Hippocampus

"a novel type of GABAergic interneuron located in the outer molecular layer (OML) of the rat dentate gyrus"

"OML interneurons were recorded in hippocampal slices by using the whole-cell patch-clamp configuration. During recording, cells were filled with biocytin for subsequent light and electron microscopic analysis. Neurons projecting to the subiculum were distributed throughout the entire OML. They had round or ovoid somata and a multipolar dendritic morphology. Two axonal domains could be distinguished: an extensive, tangential distribution within the OML and a long-range vertical and tangential projection to layer 1 and stratum pyramidale of the subiculum"

"OML interneurons were characterized physiologically by short action potential duration and marked afterhyperpolarization that followed the spike. On sustained current injection, they generated high-frequency (up to 130 Hz, 34°C) trains of action potentials with only little adaptation. In situ hybridization and single-cell RT-PCR analysis for GAD67 mRNA confirmed the GABAergic nature of OML interneurons."

(Hooft et al., 2000)

Differential Expression of Group I Metabotropic Glutamate Receptors in Functionally Distinct Hippocampal Interneurons

"Four major interneuron subtypes could be distinguished based on the mGluR-mediated inward current induced by the application of 100 μmtrans-(1S,3R)-1-aminocyclopentane-1,3-dicarboxylic acid (ACPD) under voltage-clamp conditions and the action potential firing pattern under current-clamp conditions. Type I interneurons responded with a large inward current of □ 224pA, were positive for somatostatin, and the majority expressed both mGluR1 and mGluR5. Type II interneurons responded with an inward current of □ 80pA, contained calbindin, and expressed mainly mGluR1. Type III interneurons responded with an inward current of □ 60pA. These interneurons were fast-spiking, contained parvalbumin, and expressed mainly mGluR5. Type IV interneurons did not respond with an inward current upon application of ACPD, yet they expressed group I mGluRs."

(Hosp et al., 2013)

Morpho-physiological Criteria Divide Dentatecc Gyrus Interneurons into Classes

"GABAergic inhibitory interneurons control fundamental aspects of neuronal network function"

"We identified five interneuron classes, each of them characterized by a distinct set of anatomical and physiological parameters"

"GABAergic cells fell into four morphological classes but diverged into three physiological classes with strong overlap among BC- and HIPP-like neurons in class P1 and between TML-, ML-, and HIPP-like cells in class P3"

"ML cells formed an exception; they could be classified on the basis of only their electrophysiological characteristics. TML cells could be largely but not entirely classified on the basis of electrophysiological parameters"

"a combination of morphological and physiological characteristics resulted in the emergence of five classes with strong correlations between morphologically defined interneuron types and physiological properties"

(Mott et al., 1997)

Interneurons of the Dentate–Hilus Border of the Rat Dentate Gyrus: Morphological and Electrophysiological Heterogeneity

"Interneurons located near the border of the dentate granule cell layer and the hilus were studied in hippocampal slices using whole-cell current clamp and biocytin staining. Because these interneurons exhibit both morphological and electrophysiological diversity, we asked whether passive electrotonic parameters or repetitive firing behavior correlated with axonal distribution. Each interneuron was distinguished by a preferred axonal distribution in the molecular layer or granule cell layer, and four groups could be discerned, the axons of which arborized in (1) the granule cell layer, (2) the inner molecular layer, (3) the outer molecular layer, and (4) diffusely in the molecular layer. In our sample, interneurons with axons arborizing diffusely in the molecular layer were most frequent, and

those with axons restricted to the granule cell layer were least frequent. Resting potential, input resistance, time constant, electrotonic length, and spike frequency adaptation (SFA) were not significantly different among the four groups, and the variability in SFA between cells with similar axonal distributions was striking. Clear differences in action potential morphology and afterhyperpolarizations, however, emerged when nonadapting interneurons were compared with those exhibiting SFA. Interneurons exhibiting SFA had characteristically broader spikes, progressive slowing of action potential repolarization during repetitive firing, and slow afterhyperpolarizations that distinguished them from nonadapting interneurons. We propose that the variability in repetitive firing behavior and morphology exhibited by each of these interneurons makes each interneuron unique and may provide a high level of fine tuning of inhibitory control critical to information processing in the dentate."

(Perrenoud et al., 2022)

Molecular and electrophysiological features of GABAergic neurons in the dentate gyrus reveal limited homology with cortical interneurons

"In the hilus of the dentate gyrus, different classification schemes of GABAergic interneurons have been proposed, based on morphological features, neuropeptides and calcium-binding proteins contents or electrophysiological behaviors (Houser, 2007)(Freund et al., 1998)(Lübke et al., 1998)(Mott et al., 1997)(Han et al., 1993)(Halasy et al., 1993)(Acsády et al., 2000)(Buckmaster et al., 1995)(Jinno et al., 2002)"

"At the molecular level, these classes are well defined based on the expression of calcium binding proteins such as parvalbumin (PV) and calretinin (CR), and neuropeptides like somatostatin (SOM), Neuropeptide Y (NPY) and cholecystokinin (CCK), and enzymes such as the neuronal NO synthase (NOS1)"

"Most GABAergic interneurons populating the neocortex, hippocampus and striatum share a common developmental origin (Pelkey et al., 2017)(Hu et al., 2017)(Lim et al., 2018)(Villar-Cerviño et al., 2015). Accordingly, they tend to include the same 3 major families: (1) soma and axo-axonic neurons showing fast discharge and expressing PV, (2) dendrite targeting interneurons expressing SOM and (3) neurogliaform neurons expressing NPY/ Reelin (Kawaguchi et al., 1997)[27](Tremblay et al., 2016)(Tepper et al., 2010)(Tepper et al., 2018)(Klausberger et al., 2008). Regions of the cortex and hippocampus also include soma targeting interneurons expressing CCK and interneurons expressing VIP projecting to other types of inhibitory cells (Pelkey et al., 2017)(Tremblay et al., 2016)."

(Tepper et al., 2010)

Heterogeneity and Diversity of Striatal GABAergic Interneurons

"The canonical view of striatal GABAergic interneurons has evolved over several decades of neuroanatomical/neurochemical and electrophysiological studies. From the anatomical studies, three distinct GABAergic interneuronal subtypes are generally recognized. The best-studied subtype expresses the calcium-binding protein, parvalbumin. The second best known interneuron type expresses a number of neuropeptides and enzymes, including neuropeptide Y, somatostatin, and nitric oxide synthase. The last GABAergic interneuron subtype expresses the calcium binding protein, calretinin. There is no overlap or co-localization of these three different sets of markers. The parvalbumin-immunoreactive GABAergic interneurons have been recorded in vitro and shown to exhibit a fast-spiking phenotype characterized by short duration action potentials with large and rapid spike AHPs. They often fire in a stuttering pattern of high frequency firing interrupted by periods of silence. They are capable of sustained firing rates of over 200 Hz. The NPY/SOM/NOS interneurons have been identified as PLTS cells, exhibiting very high input resistances, low threshold spike and prolonged plateau potentials in response to intracellular depolarization or excitatory synaptic stimulation. Thus far, no recordings from identified CR interneurons have been obtained. Recent advances in technological approaches, most notably the generation of several BAC transgenic mouse strains which express a fluorescent marker, enhanced green fluorescent protein, specifically and selectively only in neurons of a certain genetic makeup (e.g., parvalbumin-, neuropeptide Y-, or tyrosine hydroxylase-expressing neurons etc.) have led to the ability of electrophysiologists to visualize and patch specific neuron types in brain slices with epifluorescence illumination. This has led to a rapid expansion of the number of neurochemically and/or electrophysiologically identified interneuronal cell types in the striatum and elsewhere. This article will review the anatomy, neurochemistry, electrophysiology, synaptic connections, and function of the three "classic" striatal GABAergic interneurons as well as more recent data derived from in vitro recordings from BAC transgenic mice as well as recent in vivo data."

Transcriptomic Interneuron Classifications

Single-cell RNA sequencing has revealed much greater interneuron diversity than previously recognized, identifying 23-100 transcriptomic types in cortical and hippocampal regions. These transcriptomic classifications are based on gene expression patterns and provide molecular markers that distinguish interneuron subtypes, but researchers are still working to understand how these molecular identities relate to functional properties. (13 sources)

The advent of single-cell RNA sequencing has revolutionized interneuron classification by revealing enormous genomic diversity that extends far beyond traditional morphological and physiological categories (Ekins et al., 2020) (Olah et al., 2019) (Fuzik et al., 2015) (Que et al., 2019) (Tasic, 2018). While morphological

approaches identified 16 or more interneuron types in the hippocampus, transcriptomic cell typing has identified 23 interneuron types in the cortex (Contreras et al., 2019). Even more comprehensive analyses suggest that the number of GABAergic types may be much higher than previously recognized, with recent studies describing up to 100 transcriptomic types in hippocampal and cortical regions (Tzilivaki et al., 2023) (Yao et al., 2020).

Transcriptomic profiling has provided detailed molecular signatures for major interneuron classes. Analysis of dendritic transcriptomes in the hippocampus has identified five distinct neuron types, including three GABAergic subtypes distinguished by specific gene expression patterns (Perez et al., 2020). GABAergic type 1 neurons are characterized by expression of somatostatin (Sst) and other neuropeptides, while GABAergic type 3 neurons express cholecystokinin (Cck) and cannabinoid receptor 1 (Cnr1), and GABAergic type 2 neurons show intermediate expression patterns (Perez et al., 2020).

The transcriptomic approach has been particularly valuable for dissecting the diversity within major interneuron classes. For somatostatin-expressing interneurons alone, recent single-cell transcriptome analyses have provided a comprehensive subtype census that serves as a molecular ground truth of neuronal identity (Chamberland et al., 2023) (Chamberland et al., 2024). These studies have shown that hippocampal somatostatin-expressing interneurons can be divided into at least four subfamilies based on combinatorial gene expression patterns, with specific intersections like Sst::Tac1 targeting bistratified cells and Ndnf::Nkx2-1 targeting oriens lacunosum-moleculare interneurons (Chamberland et al., 2024).

Within the parvalbumin subclass, transcriptomic diversity is particularly striking. CA1 appears to contain at least seven and as many as 14 PV-expressing transcriptomic subtypes (Tzilivaki et al., 2023) (Yao et al., 2020). Similarly, other broad interneuron subclasses including SOM, Lamp5, Sncg, and VIP show extensive transcriptomic diversity (Tzilivaki et al., 2023).

Transcriptomic classifications have also revealed important molecular features related to synaptic function and connectivity. Many cell type-specific features involve ion channels, receptors, synaptic organizers, and trans-synaptic signaling molecules (Que et al., 2019). For example, transcriptional profiling indicates conserved expression of specific genes like Grin3a in neocortical neurogliaform cells defined by Id2 and Lamp5 expression (Kim et al., 2025).

Despite these advances, a major challenge remains in connecting transcriptomic classifications to functional properties. While these studies have accelerated cell classification efforts by identifying numerous genes that define neuronal subtypes, the functional relevance of most transcriptomic marker genes remains poorly understood (Olah et al., 2019). Current research efforts focus on correlating these transcriptomic datasets with previously identified interneuron subtypes based on morphological and physiological criteria (Ekins et al., 2020) (Lim et al., 2018). The connections between molecular identities and specific functions are not yet fully understood, and there is still much work to be done to determine how these transcriptomic cell types relate to traditional classifications based on morphological or molecular markers (Chamberland et al., 2023) (Tzilivaki et al., 2023).

Evidence

(Ekins et al., 2020)

Emergence of non-canonical parvalbumin-containing interneurons in hippocampus of a murine model of type I lissencephaly

> *"Inhibitory interneurons are classified based on a combination of their morphological, biochemical, intrinsic electrical, and connectivity properties (Lim et al., 2018). Advances in single-cell RNA sequencing have revealed enormous diversity in interneuron genomics, and current efforts attempt to correlate transcriptomic data sets with previously identified interneuron subtypes (Tasic et al., 2018)Muñoz-Manchado et al., 2018;(Gouwens et al., 2019)Que and Lukacsovich, 2020). In CA1 hippocampus alone, inhibitory synaptic transmission is mediated by at least 15 different subtypes of GABAergic inhibitory interneurons . Three canonical interneuron subtypes express the calcium-binding protein parvalbumin (PV): basket-cells, axo-axonic cells, and bistratified cells"*

> *"WT PV+INTs consist of two physiological subtypes (80% fast-spiking (FS), 20% non-fast-spiking (NFS)) and four morphological subtypes (basket, axo-axonic, bistratified, radiatum-targeting)."*

(Olah et al., 2019)

Functional specification of CCK+ interneurons by alternative isoforms of Kv4.3 auxiliary subunits

"The biophysical and morphological properties of complementary GABAergic cell classes are specifically tuned for regulating the activity of the much more populous principal cells in broad temporal (from second to sub-millisecond; (Hu et al., 2014)(Overstreet-Wadiche et al., 2015) and spatial (from axons to distal dendrites Freund and Buzsáki, 1996) domains"

"Emerging evidence using single-cell transcriptomics suggests that the number of GABAergic types may be higher than previously recognized (Földy et al., 2016)(Fuzik et al., 2015)(2017)(Que et al., 2019)(Tasic, 2018)Zeisel et al., 2015). While these studies accelerated cell classification efforts by identifying a large number of genes that appear to define further subtypes in neuronal taxonomy, the functional relevance of most transcriptomic marker genes remain poorly understood"

"Several properties distinguish CCK+INs from other major GABAergic cell classes."

(Fuzik et al., 2015)

Integration of electrophysiological recordings with single-cell RNA-seq data identifies novel neuronal subtypes

"Traditionally, neuroscientists have defined the identity of neurons by the cells' location, morphology, connectivity and excitability. However, the direct relationship between these parameters and the molecular phenotypes has remained largely unexplored. Here, we present a method for obtaining full transcriptome data from single neocortical pyramidal cells and interneurons after whole-cell patch-clamp recordings in mouse brain slices. In our approach, termed Patch-seq, a patch-clamp stimulus protocol is followed by the aspiration of the entire somatic compartment into the recording pipette, reverse transcription of RNA including addition of unique molecular identifiers, cDNA amplification, Illumina library preparation and sequencing. We show that Patch-seq reveals a close link between electrophysiological characteristics, responses to acute chemical challenges and RNA expression of neurotransmitter receptors and channels. Moreover, it distinguishes neuronal subpopulations that correspond to both well-established and, to our knowledge, hitherto undescribed neuronal subtypes. Our findings demonstrate the ability of Patch-seq to precisely map neuronal subtypes and predict their network contributions in the brain."

(Que et al., 2019)

Deep Survey of GABAergic Interneurons: Emerging Insights From Gene-Isoform Transcriptomics

"GABAergic interneuron diversity is a key feature in the brain that helps to create different brain activity patterns and behavioral states. Cell type classification schemes—based on anatomical, physiological and molecular features—have provided us with a detailed understanding of the distinct types that constitute this diversity and their contribution to brain function. Over recent years, the utility of single-cell RNAseq has majorly complemented this existing framework, vastly expanding our knowledge base, particularly regarding molecular features. Single-cell gene-expression profiles of tens of thousands of GABAergic cells from many different types are now available. The analysis of these data has shed new lights onto previous classification principles and illuminates a path towards a deeper understanding of molecular hallmarks behind interneuron diversity. A large part of such molecular features is synapse-related. These include ion channels and receptors, as well as key synaptic organizers and trans-synaptic signaling molecules. Increasing evidence suggests that transcriptional and post-transcriptional modifications further diversify these molecules and generate cell type-specific features. Thus, unraveling the cell type-specific nature of gene-isoform expression will be a key in cell type classification. This review article discusses progress in the transcriptomic survey of interneurons and insights that have begun to manifest from isoform-level analyses."

(Tasic, 2018)

Single cell transcriptomics in neuroscience: cell classification and beyond.

"Biology has been facing a daunting problem since the cell was understood to be the building block of metazoans: how do we study multicellular systems, when a universal approach to characterize their building blocks and classify them does not exist? Metazoan diversity has not helped: there are many model and non-model organisms, developmental and adult stages, healthy and diseased states. Here, I review the application of single cell transcriptomics to cell classification in neuroscience and its corollaries: the differentially expressed genes discovered in this process are a treasure trove for understanding cell type function and enabling specific access to those types. The advancements and widespread adoption of single-cell transcriptomics are bound to transform our understanding of neural system development, function, pathology and evolution."

(Contreras et al., 2019)

Molecular Specialization of GABAergic Synapses on the Soma and Axon in Cortical and Hippocampal Circuit Function and Dysfunction

"Interneurons can be classified by their morphology, connectivity, firing pattern, and gene expression pattern. Based on morphology, 16 or more types of interneuron have been distinguished in the hippocampus (Cajal, 1893;Lorente and De Nó, 1934;(Parra et al., 1998) while transcriptomic cell typing has identified 23 interneuron types in the cortex (Tasic et al., 2016)"

"the majority of interneurons in these regions express either the neuropeptide cholecystokinin or the calcium binding protein

parvalbumin (CCK; PV; (Whissell et al., 2015)."

(Tzilivaki et al., 2023)

Hippocampal GABAergic interneurons and memory

""Classic" interneuron classifications"

"CCK cells are a remarkably diverse group in the hippocampus (Cope et al., 2002)(Lasztóczi et al., 2011) (Figure 1) that extends beyond Sncg cells. (Dudok et al., 2021) The majority of cells in the VIP subclass are interneuron-specific (IS), specifically targeting other interneurons rather than PCs. Three types of IS interneurons have been described, mostly in the hippocampus (Figure 1), and they are likely to play important disinhibitory roles. (Guet-McCreight et al., 2020)"

"Freund and colleagues first characterized IS interneurons and showed that these cells express calretinin (CR) (IS-1), VIP (IS-2), or both (IS-3)"

"recent single-cell transcriptomic datasets have led to a more extensive taxonomy of hippocampal interneurons, (Dudok et al., 2021) (Yao et al., 2020)(2017)51 with up to 100 types having been described, the functional significance of these transcriptomic types remains mostly uninvestigated"

"there is still much work to be done to determine how these transcriptomic cell types relate to the more traditional classifications such as morphological or molecular markers"

"For the other broad subclasses of interneurons (SOM, Lamp5, Sncg, and VIP), less is known"

"Recent mouse transcriptomic studies indicate a huge diversity of interneurons beyond the level of the cell types discussed thus far. Focusing on the PV subclass as an example, CA1 appears to contain at least seven (2017) and as many as 14 PV-expressing transcriptomic subtypes. (Yao et al., 2020)"

"a recent study showed that in visual cortex, the morphological and electrical properties of these cells are linked to transcriptomic subtype identity. 176"

(Yao et al., 2020)

A taxonomy of transcriptomic cell types across the isocortex and hippocampal formation

"The isocortex and hippocampal formation are two major structures in the mammalian brain that play critical roles in perception, cognition, emotion and learning. Both structures contain multiple regions, for many of which the cellular composition is still poorly understood. In this study, we used two complementary single-cell RNA-sequencing approaches, SMART-Seq and 10x, to profile ▯ 1.2 million cells covering all regions in the adult mouse isocortex and hippocampal formation, and derived a cell type taxonomy comprising 379 transcriptomic types. The completeness of coverage enabled us to define gene expression variations across the entire spatial landscape without significant gaps. We found that cell types are organized in a hierarchical manner and exhibit varying degrees of discrete or continuous relatedness with each other. Such molecular relationships correlate strongly with the spatial distribution patterns of the cell types, which can be region-specific, or shared across multiple regions, or part of one or more gradients along with other cell types. Glutamatergic neuron types have much greater diversity than GABAergic neuron types, both molecularly and spatially, and they define regional identities as well as inter-region relationships. For example, we found that glutamatergic cell types between the isocortex and hippocampal formation are highly distinct from each other yet possess shared molecular signatures and corresponding layer specificities, indicating their homologous relationships. Overall, our study establishes a molecular architecture of the mammalian isocortex and hippocampal formation for the first time, and begins to shed light on its underlying relationship with the development, evolution, connectivity and function of these two brain structures."

(Perez et al., 2020)

Subcellular sequencing of single neurons reveals the dendritic transcriptome of GABAergic interneurons

"To investigate the effect of cell type on the dendritic transcriptome, we first characterized the cell-types present in our dataset using our somatic samples. Dimensionality reduction among somata revealed five neuron-types: two glutamatergic and three GABAergic (Figure 1D). Figure S2A shows the most differentially expressed genes between identified cell types (adjusted $p < 0.05$, logistic regression model). The major differences in expression occurred between glutamatergic and GABAergic somata, however, significant differences between subtypes were also observed. Although glutamatergic types 1 and 2 largely co-expressed a majority of transcripts, a handful of mRNAs like Nptx2 and Grm5 allowed their distinction (Figure 1E). GABAergic 1 neurons, the most abundant cell type in our dataset, were distinguished by their expression of Sst and other neuropeptides. Despite their clear inhibitory identity, GABAergic 3 neurons exhibited a very different transcriptomic profile from GABAergic 1, mainly distinguished by the expression of Cck and Cnr1. Finally, GABAergic 2 neurons showed similarities in their expression patterns to both GABAergic 1 and 3; their identity was best described by the co-expression of some GABAergic 3 markers, like Id2, but not others, like Cck (Figure 1E and S2A). To benchmark the breadth of cell-types detected we compared our cell-types to those previously described in single-cell studies of the hippocampus. For this we used Seurat.v3 integration algorithm (Butler et al., 2018), which identifies shared populations between different single cell datasets and projects them into a common space. We compared our somata dataset with

two CA1-derived scRNA-seq datasets. These studies sampled CA1 cells and profiled the transcriptome of 1314 CA1 cells (Zeisel et al., 2015) or 3,663 CA1 GABAergic neurons"

(Chamberland et al., 2023)

Functional specialization of hippocampal somatostatin-expressing interneurons

"Recent single cell transcriptome analyses have provided a comprehensive Sst-IN subtype census, a plausible molecular ground truth of neuronal identity whose links to specific functionality remain incomplete"

"Four mouse models based on single or combinatorial Cre- and Flp- expression differentiated functionally distinct subpopulations of CA1 hippocampal Sst-INs that largely tiled the morpho-functional parameter space of the Sst-INs superfamily"

"our results provide a framework to translate neuronal transcriptomic identity into discrete functional subtypes that capture the diverse specializations of hippocampal Sst-INs"

"Recent transcriptome analyses have provided a comprehensive classification of interneuron subtypes, but the connections between molecular identities and specific functions are not yet fully understood"

"Functional investigation in transgenic animals revealed that hippocampal somatostatin-expressing interneurons (Sst-INs) can be divided into at least four subfamilies, each with distinct functions"

(Chamberland et al., 2024)

Functional specialization of hippocampal somatostatin-expressing interneurons

"Recent transcriptome analyses have provided a comprehensive classification of interneurons, but the connections between molecular identities and specific functions are not yet fully understood"

"hippocampal somatostatin-expressing interneurons (Sst-INs) can be divided into at least four subfamilies, each with distinct functions"

"the Sst;;Tac1 intersection targeted a population of bistratified cells that overwhelmingly targeted fast-spiking interneurons. In contrast, the Ndnf;;Nkx2-1 intersection revealed a population of oriens lacunosum-moleculare interneurons that selectively targeted CA1 pyramidal cells"

"genetically distinct subfamilies of Sst-INs form specialized circuits in the hippocampus."

(Kim et al., 2025)

Discrete interneuron subsets participate in GluN1/GluN3A excitatory glycine receptor (eGlyR)-mediated regulation of hippocampal network activity throughout development and evolution

"hippocampal somatostatin and neurogliaform interneurons (Sst-INs and NGFCs)"

"parvalbumin (PV) IN"

"Grin3a is minimally expressed within transcripZonally defined rodent hippocampal and neocorZcal PV INs"

"hippocampal MGE-and CGE-derived NGFC/IvyC cohorts"

"TranscripZonal profiling indicates conserved strong Grin3a expression levels in neocorZcal NGFCs defined by Id2 and Lamp5 expression"

"Sst-INs as major contributors to early postnatal hippocampal and neocorZcal network GABAergic tone"

(Lim et al., 2018)

Development and Functional Diversification of Cortical Interneurons.

"In the cerebral cortex, GABAergic interneurons have evolved as a highly heterogeneous collection of cell types that are characterized by their unique spatial and temporal capabilities to influence neuronal circuits. Current estimates suggest that up to 50 different types of GABAergic neurons may populate the cerebral cortex, all derived from progenitor cells in the subpallium, the ventral aspect of the embryonic telencephalon. In this review, we provide an overview of the mechanisms underlying the generation of the distinct types of interneurons and their integration in cortical circuits. Interneuron diversity seems to emerge through the implementation of cell-intrinsic genetic programs in progenitor cells, which unfold over a protracted period of time until interneurons acquire mature characteristics. The developmental trajectory of interneurons is also modulated by activity-dependent, non-cell-autonomous mechanisms that influence their ability to integrate in nascent circuits and sculpt their final distribution in the adult cerebral cortex."

Molecular Markers and Nomenclature Systems

GABAergic interneurons are classified using a standardized set of molecular markers including calcium-binding proteins (parvalbumin, calretinin, calbindin), neuropeptides (somatostatin, CCK, NPY, VIP), and enzymes (neuronal nitric oxide synthase). These markers define three major developmental families: fast-spiking parvalbumin-positive cells, dendrite-targeting somatostatin-positive cells, and neurogliaform NPY/Reelin-positive cells. (15 sources)

The molecular classification of GABAergic interneurons relies on a well-established nomenclature system based on specific protein markers that define distinct interneuron populations. At the molecular level, these classes are well defined based on the expression of calcium binding proteins such as parvalbumin (PV) and calretinin (CR), and neuropeptides like somatostatin (SOM), Neuropeptide Y (NPY) and cholecystokinin (CCK), and enzymes such as the neuronal NO synthase (NOS1) (Perrenoud et al., 2022). The calcium-binding proteins and peptides that characterize different subpopulations of GABAergic interneurons show specific co-expression patterns, such as substantial co-expression of certain markers with calbindin, but not parvalbumin or calretinin (Huang et al., 2014).

The field has identified three major families of GABAergic interneurons that are conserved across cortical regions based on their molecular signatures and developmental origins. Most GABAergic interneurons include the same 3 major families: (1) soma and axo-axonic neurons showing fast discharge and expressing PV, (2) dendrite targeting interneurons expressing SOM and (3) neurogliaform neurons expressing NPY/Reelin (Perrenoud et al., 2022) (Tepper et al., 2010) (Lim et al., 2018) (Tepper et al., 2018). Additional specialized populations include soma targeting interneurons expressing CCK and interneurons expressing VIP that project to other types of inhibitory cells (Perrenoud et al., 2022).

The two most prominent molecular markers define the major GABAergic populations: parvalbumin (PV) and somatostatin (SST) interneurons, the two major types of GABAergic inhibitory interneurons in the hippocampus, exhibit distinct morpho-physiological properties (Huang et al., 2023). Within the parvalbumin population, interneurons expressing the calcium binding protein parvalbumin (PV) make up approximately 40% of all GABAergic interneurons, though this represents a heterogeneous group of functionally distinct interneuron subtypes including basket, axo-axonic and bistratified cells (Dannenberg et al., 2017).

The molecular markers also reveal important functional distinctions between interneuron subtypes. Most CCK-positive terminals surrounding pyramidal cell somata were cholecystokinin-positive (CCK) GABAergic interneurons (basket cells), while parvalbumin-positive GABAergic interneuron terminals were negative for cannabinoid CB1 receptors (Rivera et al., 2014) (Katona et al., 1999). This molecular distinction reflects functional differences, as CCK-positive basket cells are modulated by cannabinoid signaling while PV-positive basket cells are not.

Developmental origin further refines the molecular classification system. The majority of GABAergic interneurons originate from the medial ganglionic eminence (MGE) or the caudal ganglionic eminence (CGE), with different origins forming separate interneuron classes that display distinct cellular properties (Wierenga et al., 2010) (Gelman et al., 2009) (Xu et al., 2004) (Miyoshi et al., 2010). CGE-derived interneurons predominantly express either reelin (70%) or vasoactive intestinal peptide (VIP; 15%), while MGE-derived cells typically express parvalbumin and somatostatin (Wierenga et al., 2010).

Current research employs standardized marker panels to quantify interneuron populations, with studies routinely quantifying four types of GABAergic neurons: PV+, CR+, SST+, and CCK+ interneurons (Futacsi et al., 2025). These canonical interneuron types conserved across distinct cortical regions have been defined by their morphological, electrophysiological, and neurochemical properties (Ishino et al., 2017). The distinct expression profiles of neuropeptides relate to points of synaptic contact, morphological structure and intrinsic excitability, providing a molecular framework that distinguishes interneuron subtypes (Dannenberg et al., 2017).

Evidence

(Perrenoud et al., 2022)

Molecular and electrophysiological features of GABAergic neurons in the dentate gyrus reveal limited homology with cortical interneurons

"In the hilus of the dentate gyrus, different classification schemes of GABAergic interneurons have been proposed, based on morphological features, neuropeptides and calcium-binding proteins contents or electrophysiological behaviors (Houser, 2007)(Freund et al., 1998)(Lübke et al., 1998)(Mott et al., 1997)(Han et al., 1993)(Halasy et al., 1993)(Acsády et al., 2000)(Buckmaster et al., 1995)(Jinno et al., 2002)"

"At the molecular level, these classes are well defined based on the expression of calcium binding proteins such as parvalbumin (PV) and calretinin (CR), and neuropeptides like somatostatin (SOM), Neuropeptide Y (NPY) and cholecystokinin (CCK), and enzymes such as the neuronal NO synthase (NOS1)"

"Most GABAergic interneurons populating the neocortex, hippocampus and striatum share a common developmental origin (Pelkey et al., 2017)(Hu et al., 2017)(Lim et al., 2018)(Villar-Cerviño et al., 2015). Accordingly, they tend to include the same 3 major families: (1) soma and axo-axonic neurons showing fast discharge and expressing PV, (2) dendrite targeting interneurons expressing SOM and (3) neurogliaform neurons expressing NPY/ Reelin (Kawaguchi et al., 1997)[27](Tremblay et al., 2016)(Tepper et al., 2010)(Tepper et al., 2018)(Klausberger et al., 2008). Regions of the cortex and hippocampus also include soma targeting interneurons expressing CCK and interneurons expressing VIP projecting to other types of inhibitory cells (Pelkey et al., 2017)(Tremblay et al., 2016)."

(Huang et al., 2014)

Cell type-specific expression of Eps8 in the mouse hippocampus

"With respect to the calcium-binding proteins and peptides that characterize different subpopulations of GABAergic interneurons [13](Gulyás et al., 1992)(Freund et al., 1998), we found substantial co-expression of Eps8 and calbindin, but not parvalbumin or calretinin. We also found that Eps8 is present in a subset of cholecystokinin-expressing interneurons. Taken these findings into consideration, our results suggest that Eps8expressing cells in hippocampal area CA1 predominantly belong to the calbindin-and cholecystokinin-expressing subpopulations of GABAergic interneurons. In addition to using molecular expression profiles, GABAergic interneurons can be distinguished on the basis of their morphology, electrophysiological features, and innervations of distinct subcellular domains of pyramidal cells (DeFelipe, 1997)(Somogyi et al., 1998). So far, more than 20 different types of interneurons have been recognized in the hippocampus and neocortex (Bartolini et al., 2013). It has been reported previously that CA1 pyramidal cells are supported by at least 16 distinct types of GABAergic interneurons, each having highly stereotypic laminar arrangements and functions (Somogyi et al., 2005)(Klausberger et al., 2008)."

(Tepper et al., 2010)

Heterogeneity and Diversity of Striatal GABAergic Interneurons

"The canonical view of striatal GABAergic interneurons has evolved over several decades of neuroanatomical/neurochemical and electrophysiological studies. From the anatomical studies, three distinct GABAergic interneuronal subtypes are generally recognized. The best-studied subtype expresses the calcium-binding protein, parvalbumin. The second best known interneuron type expresses a number of neuropeptides and enzymes, including neuropeptide Y, somatostatin, and nitric oxide synthase. The last GABAergic interneuron subtype expresses the calcium binding protein, calretinin. There is no overlap or co-localization of these three different sets of markers. The parvalbumin-immunoreactive GABAergic interneurons have been recorded in vitro and shown to exhibit a fast-spiking phenotype characterized by short duration action potentials with large and rapid spike AHPs. They often fire in a stuttering pattern of high frequency firing interrupted by periods of silence. They are capable of sustained firing rates of over 200 Hz. The NPY/SOM/NOS interneurons have been identified as PLTS cells, exhibiting very high input resistances, low threshold spike and prolonged plateau potentials in response to intracellular depolarization or excitatory synaptic stimulation. Thus far, no recordings from identified CR interneurons have been obtained. Recent advances in technological approaches, most notably the generation of several BAC transgenic mouse strains which express a fluorescent marker, enhanced green fluorescent protein, specifically and selectively only in neurons of a certain genetic makeup (e.g., parvalbumin-, neuropeptide Y-, or tyrosine hydroxylase-expressing neurons etc.) have led to the ability of electrophysiologists to visualize and patch specific neuron types in brain slices with epifluorescence illumination. This has led to a rapid expansion of the number of neurochemically and/or electrophysiologically identified interneuronal cell types in the striatum and elsewhere. This article will review the anatomy, neurochemistry, electrophysiology, synaptic connections, and function of the three "classic" striatal GABAergic interneurons as well as more recent data derived from in vitro recordings from BAC transgenic mice as well as recent in vivo data."

(Lim et al., 2018)

Development and Functional Diversification of Cortical Interneurons.

"In the cerebral cortex, GABAergic interneurons have evolved as a highly heterogeneous collection of cell types that are characterized by their unique spatial and temporal capabilities to influence neuronal circuits. Current estimates suggest that up to 50 different types of GABAergic neurons may populate the cerebral cortex, all derived from progenitor cells in the subpallium, the ventral aspect of the embryonic telencephalon. In this review, we provide an overview of the mechanisms underlying the generation

of the distinct types of interneurons and their integration in cortical circuits. Interneuron diversity seems to emerge through the implementation of cell-intrinsic genetic programs in progenitor cells, which unfold over a protracted period of time until interneurons acquire mature characteristics. The developmental trajectory of interneurons is also modulated by activity-dependent, non-cell-autonomous mechanisms that influence their ability to integrate in nascent circuits and sculpt their final distribution in the adult cerebral cortex."

(Tepper et al., 2018)

Heterogeneity and Diversity of Striatal GABAergic Interneurons: Update 2018

"Our original review, "Heterogeneity and Diversity of Striatal GABAergic Interneurons," to which this is an invited update, was published in December, 2010 in Frontiers is Neuroanatomy. In that article, we reviewed several decades' worth of anatomical and electrophysiological data on striatal parvalbumin (PV)-, neuropeptide Y (NPY)- and calretinin(CR)-expressing GABAergic interneurons from many laboratories including our own. In addition, we reported on a recently discovered novel tyrosine hydroxylase (TH) expressing GABAergic interneuron class first revealed in transgenic TH EGFP reporter mouse line. In this review, we report on further advances in the understanding of the functional properties of previously reported striatal GABAergic interneurons and their synaptic connections. With the application of new transgenic fluorescent reporter and Cre-driver/reporter lines, plus optogenetic, chemogenetic and viral transduction methods, several additional subtypes of novel striatal GABAergic interneurons have been discovered, as well as the synaptic networks in which they are embedded. These findings make it clear that previous hypotheses in which striatal GABAergic interneurons modulate and/or control the firing of spiny neurons principally by simple feedforward and/or feedback inhibition are at best incomplete. A more accurate picture is one in which there are highly selective and specific afferent inputs, synaptic connections between different interneuron subtypes and spiny neurons and among different GABAergic interneurons that result in the formation of functional networks and ensembles of spiny neurons."

(Huang et al., 2023)

Differential expression of GABAA receptor subunits δ and α6 mediates tonic inhibition in parvalbumin and somatostatin interneurons in the mouse hippocampus

"Parvalbumin (PV) and somatostatin (SST) interneurons, the two major types of GABAergic inhibitory interneurons in the hippocampus, exhibit distinct morpho-physiological properties and coordinate information processing and memory formation."

(Dannenberg et al., 2017)

Modulation of Hippocampal Circuits by Muscarinic and Nicotinic Receptors

"Interneurons display an immense functional, morphological and genetic diversity, as described in previous reviews of the classification of interneuron subtypes (McBain et al., 2001)(Somogyi et al., 2005)(Rudy et al., 2011)(Kepecs et al., 2014)"

"Distinct expression profiles of neuropeptides that relate to points of synaptic contact, morphological structure and intrinsic excitability (Freund et al., 1998)(Kepecs et al., 2014) distinguish interneurons"

"Interneurons expressing the calcium binding protein parvalbumin (PV) make up approximately 40% of all GABAergic interneurons. However, this is a heterogeneous group of functionally distinct interneuron subtypes. For example, in the hippocampus alone, there are at least three functionally and morphologically distinct populations of PV + expressing interneurons, namely basket, axo-axonic and bistratified cells. Fast spiking interneurons in the hippocampus and neocortex are often PV + positive and target the soma of pyramidal cells."

(Rivera et al., 2014)

Localization of the cannabinoid CB1 receptor and the 2-AG synthesizing (DAGLα) and degrading (MAGL, FAAH) enzymes in cells expressing the Ca2+-binding proteins calbindin, calretinin, and parvalbumin in the adult rat hippocampus

"Parvalbumin + cells were specifically localized in the granular and polymorphic cell layers of the dentate gyrus and the strata oriens and pyramidale in CA1/3 fields of the rat hippocampus (Kosaka et al., 1987). They have been considered a subpopulation of GABAergic interneurons, including basket and axo-axonic cell types, which innervate the somata and proximal axons of pyramidal cells, respectively (Soriano et al., 1990)"

"Most CB + 1 terminals surrounding the somata and proximal dendrites of pyramidal neurons were cholecystokinin + (CCK) GABAergic interneurons (basket cells) and, to a lower extent, calbindin D-28k + GABAergic interneurons (Katona et al., 1999) (Marsicano et al., 1999)(Tsou et al., 1999). However, parvalbumin + GABAergic interneuron terminals localized in pyramidal cell layers were negative for CB 1 (Katona et al., 1999)(Marsicano et al., 1999)"

"most fiber terminals containing CB 1 /CCK probably innervate both calbindin + pyramidal neurons (probably including calbindin + / FAAH + pyramidal neurons in CA1) and calretinin + and parvalbumin + interneurons localized in the principal layers of the hippocampus (Gulyás et al., 1991)(Marsicano et al., 1999). Moreover, parvalbumin + fiber terminals, which represent another GABAergic interneuron type, could specifically innervate FAAH + /calbindin − pyramidal neurons in CA3."

(Katona et al., 1999)

Presynaptically Located CB1 Cannabinoid Receptors Regulate GABA Release from Axon Terminals of Specific Hippocampal Interneurons

"To understand the functional significance and mechanisms of action in the CNS of endogenous and exogenous cannabinoids, it is crucial to identify the neural elements that serve as the structural substrate of these actions. We used a recently developed antibody against the CB1 cannabinoid receptor to study this question in hippocampal networks. Interneurons with features typical of basket cells showed a selective, intense staining for CB1 in all hippocampal subfields and layers. Most of them (85.6%) contained cholecystokinin (CCK), which corresponded to 96.9% of all CCK-positive interneurons, whereas only 4.6% of the parvalbumin (PV)-containing basket cells expressed CB1. Accordingly, electron microscopy revealed that CB1-immunoreactive axon terminals of CCK-containing basket cells surrounded the somata and proximal dendrites of pyramidal neurons, whereas PV-positive basket cell terminals in similar locations were negative for CB1. The synthetic cannabinoid agonist WIN 55,212-2 (0.01–3 μm) reduced dose-dependently the electrical field stimulation-induced [3H]GABA release from superfused hippocampal slices, with an EC50 value of 0.041 μm. Inhibition of GABA release by WIN 55,212-2 was not mediated by inhibition of glutamatergic transmission because the WIN 55,212-2 effect was not reduced by the glutamate blockers AP5 and CNQX. In contrast, the CB1 cannabinoid receptor antagonist SR 141716A (1 μm) prevented this effect, whereas by itself it did not change the outflow of [3H]GABA. These results suggest that cannabinoid-mediated modulation of hippocampal interneuron networks operate largely via presynaptic receptors on CCK-immunoreactive basket cell terminals. Reduction of GABA release from these terminals is the likely mechanism by which both endogenous and exogenous CB1 ligands interfere with hippocampal network oscillations and associated cognitive functions."

(Wierenga et al., 2010)

Molecular and Electrophysiological Characterization of GFP-Expressing CA1 Interneurons in GAD65-GFP Mice

"GABAergic interneurons comprise ,10-20% of the total neuronal population and are essential for controlling and synchronizing the output of the principal cells (Danglot et al., 2006)[2](Klausberger et al., 2008). There are many different types of interneurons, executing diverse functions in shaping the activity of neuronal networks. It has proven difficult to formulate an unequivocal definition of the different interneuron types that exist in the brain (Ascoli et al., 2008)"

"The majority of GABAergic interneurons originate from the medial ganglionic eminence (MGE) or the caudal ganglionic eminence (CGE) (Miyoshi et al., 2010)(Nery et al., 2002)(Xu et al., 2004). In addition, a small fraction of interneurons are generated in the preoptic area (Gelman et al., 2010)(Gelman et al., 2009). Interneurons with different origin form separate interneuron classes and display distinct cellular properties. A full understanding of the developmental relationship between different types of interneurons will greatly contribute to define an unambiguous interneuron classification"

"The majority of them expresses either reelin (70±2%) or vasoactive intestinal peptide (VIP; 15±2%), while expression of parvalbumin and somatostatin is virtually absent. This strongly suggests they originate from the caudal, and not the medial, ganglionic eminence. GFP-labeled interneurons can be subdivided according to the (partially overlapping) expression of neuropeptide Y (42±3%), cholecystokinin (25±3%), calbindin (20±2%) or calretinin (20±2%)."

(Gelman et al., 2009)

The Embryonic Preoptic Area Is a Novel Source of Cortical GABAergic Interneurons

"GABA-containing (GABAergic) interneurons play an important role in the function of the cerebral cortex. Through mostly inhibitory mechanisms, interneurons control hyperexcitability and synchronize and shape the spatiotemporal dynamics of cortical activity underlying various brain functions. Studies over the past 10 years have demonstrated that, in most mammals, interneurons originate during development from the subcortical telencephalon—the subpallium—and reach the cerebral cortex through tangential migration. Until now, interneurons have been demonstrated to derive exclusively from two subpallial regions, the medial ganglionic eminence and the caudal ganglionic eminence. Here, we show that another subpallial structure, the preoptic area, is a novel source of cortical GABAergic interneurons in the mouse. In utero labeling and genetic lineage-tracing experiments demonstrate that neurons born in this region migrate to the neocortex and hippocampus, where they differentiate into a distinct population of GABAergic interneurons with relatively uniform neurochemical, morphological, and electrophysiological properties."

(Xu et al., 2004)

Origins of Cortical Interneuron Subtypes

"Cerebral cortical functions are conducted by two general classes of neurons: glutamatergic projection neurons and GABAergic interneurons. Distinct interneuron subtypes serve distinct roles in modulating cortical activity and can be differentially affected in cortical diseases, but little is known about the mechanisms for generating their diversity. Recent evidence suggests that many cortical interneurons originate within the subcortical telencephalon and then migrate tangentially into the overlying cortex. To test the hypothesis that distinct interneuron subtypes are derived from distinct telencephalic subdivisions, we have used an in vitro assay to assess the developmental potential of subregions of the telencephalic proliferative zone (PZ) to give rise to neurochemically defined interneuron subgroups. PZ cells from GFP+ donor mouse embryos were transplanted onto neonatal cortical feeder cells and assessed for their ability to generate specific interneuron subtypes. Our results suggest that the parvalbumin- and the somatostatin-expressing interneuron subgroups originate primarily within the medial ganglionic eminence (MGE) of the subcortical telencephalon, whereas the calretinin-expressing interneurons appear to derive mainly from the caudal

ganglionic eminence (CGE). These results are supported by findings from primary cultures of cortex from Nkx2.1 mutants, in which normal MGE fails to form but in which the CGE is less affected. In these cultures, parvalbumin- and somatostatin-expressing cells are absent, although calretinin-expressing interneurons are present. Interestingly, calretinin-expressing bipolar interneurons were nearly absent from cortical cultures of Dlx1/2 mutants. By establishing spatial differences in the origins of interneuron subtypes, these studies lay the groundwork for elucidating the molecular bases for their distinct differentiation pathways."

(Miyoshi et al., 2010)

Genetic Fate Mapping Reveals That the Caudal Ganglionic Eminence Produces a Large and Diverse Population of Superficial Cortical Interneurons

"By combining an inducible genetic fate mapping strategy with electrophysiological analysis, we have systematically characterized the populations of cortical GABAergic interneurons that originate from the caudal ganglionic eminence (CGE). Interestingly, compared with medial ganglionic eminence (MGE)-derived cortical interneuron populations, the initiation [embryonic day 12.5 (E12.5)] and peak production (E16.5) of interneurons from this embryonic structure occurs 3 d later in development. Moreover, unlike either pyramidal cells or MGE-derived cortical interneurons, CGE-derived interneurons do not integrate into the cortex in an inside-out manner but preferentially (75%) occupy superficial cortical layers independent of birthdate. In contrast to previous estimates, CGE-derived interneurons are both considerably greater in number (≈ 30% of all cortical interneurons) and diversity (comprised by at least nine distinct subtypes). Furthermore, we found that a large proportion of CGE-derived interneurons, including the neurogliaform subtype, express the glycoprotein Reelin. In fact, most CGE-derived cortical interneurons express either Reelin or vasoactive intestinal polypeptide. Thus, in conjunction with previous studies, we have now determined the spatial and temporal origins of the vast majority of cortical interneuron subtypes."

(Futacsi et al., 2025)

Quantification and correlation of amyloid-β plaque load, glial activation, GABAergic interneuron numbers, and cognitive decline in the young TgF344-AD rat model of Alzheimer's disease

"we have quantified the number of four types of GABAergic neurons: PV+, CR+, SST+, and CCK+ interneurons"

(Ishino et al., 2017)

Regional Cellular Environment Shapes Phenotypic Variations of Hippocampal and Neocortical Chandelier Cells

"Cortical circuits include diverse types of GABAergic interneurons (INs) that shape activity of excitatory principal neurons (PNs). Canonical IN types conserved across distinct cortical regions have been defined by their morphological, electrophysiological, and neurochemical properties."

Mappings Between Transcriptomic and Classical Classifications

Researchers are actively working to map the enormous diversity revealed by single-cell RNA sequencing (up to 100 transcriptomic types) onto traditional morphological and physiological classifications. While some clear mappings have been established, such as specific genetic markers targeting bistratified cells and oriens lacunosum-moleculare cells, the functional significance of most transcriptomic subtypes remains unknown and represents a major ongoing challenge in the field. (7 sources)

The integration of transcriptomic classifications with traditional morphological and physiological interneuron types represents one of the most significant current challenges in interneuron research. Current efforts attempt to correlate transcriptomic datasets with previously identified interneuron subtypes based on morphological, biochemical, intrinsic electrical, and connectivity properties (Ekins et al., 2020) (Lim et al., 2018). This mapping challenge has become increasingly complex as recent single-cell transcriptomic datasets have led to a more extensive taxonomy of hippocampal interneurons, with up to 100 types having been described, yet the functional significance of these transcriptomic types remains mostly uninvestigated (Tzilivaki et al., 2023) (Dudok et al., 2021) .

Several successful mapping efforts have demonstrated the potential for connecting transcriptomic identity to classical interneuron types. Within the somatostatin-expressing population, recent single cell transcriptome analyses have provided a comprehensive subtype census that serves as a molecular ground truth of neuronal identity, though links to specific functionality remain incomplete (Chamberland et al., 2023). Functional investigation using transgenic animals has revealed that hippocampal somatostatin-expressing interneurons can be divided into at least four subfamilies, each with distinct functions (Chamberland et al., 2023) (Chamberland et al., 2024). Specific genetic intersections have successfully targeted classical morphological types: the Sst::Tac1 intersection targeted a population of bistratified cells that overwhelmingly targeted fast-

spiking interneurons, while the Ndnf::Nkx2-1 intersection revealed a population of oriens lacunosum-moleculare interneurons that selectively targeted CA1 pyramidal cells (Chamberland et al., 2024).

The challenge of mapping transcriptomic diversity is particularly evident within the parvalbumin subclass, where CA1 appears to contain at least seven and as many as 14 PV-expressing transcriptomic subtypes (Tzilivaki et al., 2023) . Classical morphological and physiological classification recognizes that PV+ interneurons consist of two physiological subtypes (80% fast-spiking, 20% non-fast-spiking) and four morphological subtypes (basket, axo-axonic, bistratified, radiatum-targeting) (Ekins et al., 2020). However, the relationship between this extensive transcriptomic diversity and the established morpho-physiological categories remains largely unresolved.

Studies using dendritic transcriptome profiling have provided additional insights into the molecular basis of classical interneuron types. Analysis of hippocampal samples identified three GABAergic subtypes with distinct gene expression patterns: GABAergic type 1 neurons characterized by expression of somatostatin and other neuropeptides, GABAergic type 3 neurons expressing cholecystokinin and cannabinoid receptor 1, and GABAergic type 2 neurons showing intermediate expression patterns (Perez et al., 2020). These transcriptomic profiles align well with classical CCK+ and SST+ interneuron classifications.

Despite these advances, major gaps remain in understanding the functional relevance of transcriptomic diversity. Recent transcriptome analyses have provided a comprehensive classification of interneurons, but the connections between molecular identities and specific functions are not yet fully understood (Chamberland et al., 2024). There is still much work to be done to determine how these transcriptomic cell types relate to the more traditional classifications based on morphological or molecular markers (Tzilivaki et al., 2023). The field continues to work toward establishing comprehensive mappings that will bridge the gap between the molecular ground truth provided by transcriptomics and the functional understanding derived from classical morphological and physiological approaches.

Evidence

(Ekins et al., 2020)

Emergence of non-canonical parvalbumin-containing interneurons in hippocampus of a murine model of type I lissencephaly

> *"Inhibitory interneurons are classified based on a combination of their morphological, biochemical, intrinsic electrical, and connectivity properties (Lim et al., 2018). Advances in single-cell RNA sequencing have revealed enormous diversity in interneuron genomics, and current efforts attempt to correlate transcriptomic data sets with previously identified interneuron subtypes (Tasic et al., 2018)Muñoz-Manchado et al., 2018;(Gouwens et al., 2019)Que and Lukacsovich, 2020). In CA1 hippocampus alone, inhibitory synaptic transmission is mediated by at least 15 different subtypes of GABAergic inhibitory interneurons . Three canonical interneuron subtypes express the calcium-binding protein parvalbumin (PV): basket-cells, axo-axonic cells, and bistratified cells"*

> *"WT PV+INTs consist of two physiological subtypes (80% fast-spiking (FS), 20% non-fast-spiking (NFS)) and four morphological subtypes (basket, axo-axonic, bistratified, radiatum-targeting)."*

(Lim et al., 2018)

Development and Functional Diversification of Cortical Interneurons.

> *"In the cerebral cortex, GABAergic interneurons have evolved as a highly heterogeneous collection of cell types that are characterized by their unique spatial and temporal capabilities to influence neuronal circuits. Current estimates suggest that up to 50 different types of GABAergic neurons may populate the cerebral cortex, all derived from progenitor cells in the subpallium, the ventral aspect of the embryonic telencephalon. In this review, we provide an overview of the mechanisms underlying the generation of the distinct types of interneurons and their integration in cortical circuits. Interneuron diversity seems to emerge through the implementation of cell-intrinsic genetic programs in progenitor cells, which unfold over a protracted period of time until interneurons acquire mature characteristics. The developmental trajectory of interneurons is also modulated by activity-dependent, non-cell-autonomous mechanisms that influence their ability to integrate in nascent circuits and sculpt their final distribution in the adult cerebral cortex."*

(Tzilivaki et al., 2023)

Hippocampal GABAergic interneurons and memory

""Classic" interneuron classifications"

"CCK cells are a remarkably diverse group in the hippocampus (Cope et al., 2002)(Lasztóczi et al., 2011) (Figure 1) that extends beyond Sncg cells. (Dudok et al., 2021) The majority of cells in the VIP subclass are interneuron-specific (IS), specifically targeting other interneurons rather than PCs. Three types of IS interneurons have been described, mostly in the hippocampus (Figure 1), and they are likely to play important disinhibitory roles. (Guet-McCreight et al., 2020)"

"Freund and colleagues first characterized IS interneurons and showed that these cells express calretinin (CR) (IS-1), VIP (IS-2), or both (IS-3)"

"recent single-cell transcriptomic datasets have led to a more extensive taxonomy of hippocampal interneurons, (Dudok et al., 2021) (Yao et al., 2020)(2017)51 with up to 100 types having been described, the functional significance of these transcriptomic types remains mostly uninvestigated"

"there is still much work to be done to determine how these transcriptomic cell types relate to the more traditional classifications such as morphological or molecular markers"

"For the other broad subclasses of interneurons (SOM, Lamp5, Sncg, and VIP), less is known"

"Recent mouse transcriptomic studies indicate a huge diversity of interneurons beyond the level of the cell types discussed thus far. Focusing on the PV subclass as an example, CA1 appears to contain at least seven (2017) and as many as 14 PV-expressing transcriptomic subtypes. (Yao et al., 2020)"

"a recent study showed that in visual cortex, the morphological and electrical properties of these cells are linked to transcriptomic subtype identity. 176"

(Dudok et al., 2021)

Alternating sources of perisomatic inhibition during behavior.

"Interneurons expressing cholecystokinin (CCK) and parvalbumin (PV) constitute two key GABAergic controllers of hippocampal pyramidal cell output. Although the temporally precise and millisecond-scale inhibitory regulation of neuronal ensembles delivered by PV interneurons is well established, the in vivo recruitment patterns of CCK-expressing basket cell (BC) populations has remained unknown. We show in the CA1 of the mouse hippocampus that the activity of CCK BCs inversely scales with both PV and pyramidal cell activity at the behaviorally relevant timescales of seconds. Intervention experiments indicated that the inverse coupling of CCK and PV GABAergic systems arises through a mechanism involving powerful inhibitory control of CCK BCs by PV cells. The tightly coupled complementarity of two key microcircuit regulatory modules demonstrates a novel form of brain-state-specific segregation of inhibition during spontaneous behavior."

(Chamberland et al., 2023)

Functional specialization of hippocampal somatostatin-expressing interneurons

"Recent single cell transcriptome analyses have provided a comprehensive Sst-IN subtype census, a plausible molecular ground truth of neuronal identity whose links to specific functionality remain incomplete"

"Four mouse models based on single or combinatorial Cre- and Flp- expression differentiated functionally distinct subpopulations of CA1 hippocampal Sst-INs that largely tiled the morpho-functional parameter space of the Sst-INs superfamily"

"our results provide a framework to translate neuronal transcriptomic identity into discrete functional subtypes that capture the diverse specializations of hippocampal Sst-INs"

"Recent transcriptome analyses have provided a comprehensive classification of interneuron subtypes, but the connections between molecular identities and specific functions are not yet fully understood"

"Functional investigation in transgenic animals revealed that hippocampal somatostatin-expressing interneurons (Sst-INs) can be divided into at least four subfamilies, each with distinct functions"

(Chamberland et al., 2024)

Functional specialization of hippocampal somatostatin-expressing interneurons

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"the Sst;;Tac1 intersection targeted a population of bistratified cells that overwhelmingly targeted fast-spiking interneurons. In contrast, the Ndnf;;Nkx2-1 intersection revealed a population of oriens lacunosum-moleculare interneurons that selectively targeted

CA1 pyramidal cells"

"genetically distinct subfamilies of Sst-INs form specialized circuits in the hippocampus."

(Perez et al., 2020)

Subcellular sequencing of single neurons reveals the dendritic transcriptome of GABAergic interneurons

"To investigate the effect of cell type on the dendritic transcriptome, we first characterized the cell-types present in our dataset using our somatic samples. Dimensionality reduction among somata revealed five neuron-types: two glutamatergic and three GABAergic (Figure 1D). Figure S2A shows the most differentially expressed genes between identified cell types (adjusted p < 0.05, logistic regression model). The major differences in expression occurred between glutamatergic and GABAergic somata, however, significant differences between subtypes were also observed. Although glutamatergic types 1 and 2 largely co-expressed a majority of transcripts, a handful of mRNAs like Nptx2 and Grm5 allowed their distinction (Figure 1E). GABAergic 1 neurons, the most abundant cell type in our dataset, were distinguished by their expression of Sst and other neuropeptides. Despite their clear inhibitory identity, GABAergic 3 neurons exhibited a very different transcriptomic profile from GABAergic 1, mainly distinguished by the expression of Cck and Cnr1. Finally, GABAergic 2 neurons showed similarities in their expression patterns to both GABAergic 1 and 3; their identity was best described by the co-expression of some GABAergic 3 markers, like Id2, but not others, like Cck (Figure 1E and S2A). To benchmark the breadth of cell-types detected we compared our cell-types to those previously described in single-cell studies of the hippocampus. For this we used Seurat.v3 integration algorithm (Butler et al., 2018), which identifies shared populations between different single cell datasets and projects them into a common space. We compared our somata dataset with two CA1-derived scRNA-seq datasets. These studies sampled CA1 cells and profiled the transcriptome of 1314 CA1 cells (Zeisel et al., 2015) or 3,663 CA1 GABAergic neurons"